Retail Sales & Profitability Analysis

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1. Executive Summary:

This report presents a comprehensive financial analysis of a synthetic retail superstore dataset, consisting of 9,994 transaction records spanning from 2015 to 2018. While the company operates at a baseline profit margin of 12.5%, the analysis uncovered severe regional and categorical financial drains. By transitioning from descriptive statistics to prescriptive data modeling, the root cause of these losses was identified: an aggressive, uncapped promotional strategy. Discounts exceeding 20% systematically result in negative margins. A rigorous sensitivity analysis was conducted to model a financial recovery scenario. By simulating a corporate policy that caps promotional discounts at 20%, we project that even in the most catastrophic assumption — where the business loses the entirety of every order containing a capped item — net profit would still increase by +39.6%, driving the overall company margin up to a robust 22.8%.

2. Data Processing & Engineering:

To ensure analytical accuracy, a robust ETL (Extract, Transform, Load) pipeline was built using Python (Pandas) and PostgreSQL.

• Data Cleansing: Standardized schema structures and imputed missing geographic identifiers, specifically mapping isolated null values to Burlington, Vermont's '05401' ZIP code to preserve geographic integrity.

• Data Integrity: The Returns table (800 rows) was deduplicated to order grain (296 unique orders) before joining, preventing join fan-out from multiplying returned lines. The 16 remaining duplicate line items were identified and preserved, documenting them as legitimate split shipments rather than erroneous data entries.

3. Historical Financial Performance:

The company has demonstrated sustained year-over-year (YoY) growth. By 2018, total revenue reached $733,215.26, generating a profit of $93,439.27 and achieving a 12.74% profit margin. While the top-line growth is positive (a 20.36% YoY revenue increase in 2018), digging deeper into the data reveals critical inefficiencies.

4. Exploratory Data Analysis: Identifying the Leakage

A segmented approach was utilized to isolate performance drivers across products and geographies.

• Product Profitability: The Technology category drives the business, with Copiers ($55,617.82 profit) and Phones ($44,515.73 profit) leading the portfolio. Conversely, the Furniture category is highly problematic. Tables alone generated a net loss of -$17,725.48, alongside Bookcases which lost -$3,472.56.

• Geographic Performance: The losses are heavily concentrated in specific regions. The state of Texas is the largest financial drain, operating at a net loss of -$25,729.00. Ohio (-$16,971.00), Pennsylvania (-$15,560.00), and Illinois (-$12,608.00) follow closely behind.

5. Root Cause Analysis: The Pricing Strategy Flaw:

Are the losses in Texas and Illinois due to logistical failures, or pricing failures? By aggregating financial metrics by discount brackets, the data points to a clear pricing flaw:

• Transactions with zero discounts maintain a healthy 29.51% profit margin.

• The "tipping point" is 20%. Discounts up to this threshold remain profitable (11.82% margin).

• Beyond 20%, the unit economics collapse. A 30% discount yields a -10.05% margin, while a 40% discount deepens the loss to a -19.81% margin.

Cross-referencing this finding with our geographic and categorical losers confirmed the hypothesis. Texas and Illinois are failing because their average applied discounts are 37.02% and 39.00%, respectively. Similarly, the heavily underperforming Tables sub-category suffers from an average discount of 26.13%. The company is effectively paying customers to take these products.

6. Prescriptive Analysis & Financial Optimization:

To provide actionable business value, a sensitivity analysis was engineered in SQL to simulate capping all discounts at a maximum of 20% for loss-making transactions. Recognizing that reducing discounts will inevitably affect demand elasticity, the model tests multiple "Lost Sales" scenarios against the baseline profit of $286,397.

• The Optimistic Scenario (0% Demand Loss): If sales volume remains constant, capping discounts would increase profits by +73.5%, yielding $496,944. (Not recommended for realistic forecasting).

• The Pessimistic Scenario (100% Demand Loss): If every customer refuses to buy the corrected items, simply rejecting those unprofitable sales increases the company's net profit by +48.4%.

• The "Collateral Damage" Scenario (Full Order Loss): In retail, customers often bundle items. If a customer abandons their entire shopping cart because we capped the discount on a single item, we lose profitable items alongside the unprofitable ones. Even under this extreme worst-case scenario, the business's net profit increases by +39.6% (to $399,683) and the global profit margin improves by 10.3 percentage points, reaching 22.8%.

7. Strategic Recommendations

Based on the data, the following immediate actions are recommended:

- Implement a hard cap of 20% on all promotional discounts, particularly in the Central and East regions (Texas, Ohio, Illinois, Pennsylvania).
- Review vendor contracts for Tables and Bookcases. If a 20% discount prevents these items from selling, their base costs are too high to support a competitive retail strategy.
- Restructure regional manager KPIs. Shift the primary performance metric in underperforming states from "Total Sales Volume" to "Net Profit Margin" to disincentivize the abuse of discounts.